Jishnu Prasad

+91 98447 78936 | thisisjishnuprasad888@gmail.com | linkedin.com/in/jishnu-prasad-ba90a9328 |
github.com/Jishnu-Prasad888 | jishnu-prasad.vercel.app

Education

National Institute of Technology Karnataka — Surathkal, India
B.Tech in Electrical and Electronics Engineering CGPA: ***7.18 / 10*** — *Sep. 2024 – Present*

Experience

SPIRE (Signal Processing, Interpretation and REpresentation) Lab — IISC
Research Intern | RAG, Python, React, Agentic AI — *May 2026 – Present*

- Built a multilingual (English & Kannada) multimodal RAG system for SBI banking forms and documentation, integrating OCR, automated web scraping, document ingestion, embedding generation, and Elasticsearch-based indexing to enable hybrid vector and keyword search with metadata filtering for accurate and contextually relevant retrieval across digital and scanned documents.
- Architected a multi-agent retrieval framework with specialized agents for PDF, Markdown, Excel, and HTML documents, dynamically routing queries to the most relevant knowledge source for improved retrieval accuracy and contextual grounding.
- Designed and implemented a Knowledge-Augmented Generation (KAG) architecture to overcome limitations of traditional RAG on complex banking queries, enhancing multi-hop reasoning, contextual understanding, and answer completeness.
- Fine-tuned a Qwen3-8B-Instruct model on a custom evaluation dataset to act as an answer-quality critic, scoring responses on Accuracy, Answer Relevancy, and Faithfulness and enabling a DeepThink feedback loop that triggered iterative retrieval-generation cycles; benchmarked RAG, Multi-Agent RAG, GPT-4o, and KAG using an LLM-as-a-Judge evaluation framework.

HALE (Healthcare Analytics and Language Engineering) Lab — NITK
Research Intern | Kubernetes, Redis, LangChain, RAG, Ansible — *Jan 2026 – Present*

- Engineered a multi-node Raspberry Pi k3s cluster with MetalLB and NGINX Ingress, enabling load-balanced, TLS-secured service delivery.
- Built a full observability stack using Prometheus, Grafana, Loki, and Promtail for centralized monitoring and log aggregation.
- Deployed distributed MinIO storage, Flask/React applications, and JupyterHub, addressing multi-node storage, image distribution, and service discovery challenges.
- Automated provisioning and orchestration with Ansible to support reproducible edge inference workloads on a Raspberry Pi 5 Kubernetes cluster.

Urban Records — Remote
Freelance Full-Stack Engineer | Django REST Framework, React, TailwindCSS, Docker, PostgreSQL *Sep. 2025 – Nov. 2025*

- Architected and developed a containerized full-stack document management platform for tax, passport, and visa processing workflows using Django REST Framework, React, PostgreSQL, and Docker.
- Optimized document processing pipelines to reduce manual intervention from days to just hours
- Deployed a production-grade system supporting 500+ active users with high reliability.

SENA NGO — Surathkal, India
Backend Engineering Intern | Python, Django, SQL, Docker — *May 2025 – Jul. 2025*

- Developed and integrated a secure donations pipeline using the Cashfree Python SDK, enabling reliable online payment processing and transaction tracking.
- Built form-driven publishing workflows allowing non-technical contributors to create and publish content without developer intervention, reducing operational dependency on the engineering team .

Projects

Memora — Redis-like In-Memory Database | *Go, Python, TCP, Concurrent Systems*
https://github.com/Jishnu-Prasad888/Memora

- Engineered a Redis-compatible in-memory key-value database in Go with full RESP protocol support and concurrent TCP client handling.
- Implemented custom hash-table storage, TTL expiration, background cleanup, and RDB-style snapshot persistence for durable in-memory data management.
- Built CLI and Python GUI clients for database administration, monitoring, and interactive command execution.
- Achieved **226K+ operations/sec** with **1M benchmarked operations**, **39µs P50 latency**, and **2.0ms P99 latency** under concurrent workloads.
- Designed goroutine-based request processing and non-blocking I/O architecture to support scalable multi-client access.

ATLAS & Beacon — Distributed Fleet Management and Observability Platform | *ReactJS, Django, NATS JetStream, RAG, GraphQL, Aurora PostgreSQL, Redis, Docker, Cloudflare, Vercel*
https://github.com/Jishnu-Prasad888/ATLAS

- Engineered and deployed distributed control-plane architecture comprising centralized **ATLAS** services and lightweight **Beacon** agents, enabling **scalable fleet orchestration** and **remote operations**.
- Built a high-throughput observability pipeline collecting **CPU, memory, disk, network, process, Docker, and Kubernetes telemetry**, storing historical metrics in **Aurora PostgreSQL** and **Redis**, with real-time aggregation, log indexing, and **WebSocket-based streaming dashboards**.
- Implemented fault-tolerant agent communication over **NATS JetStream** using **durable consumers**, **message replay**, **offline buffering**, **dead-letter queues** and **mTLS**-based authentication for reliable distributed messaging.
- Developed a multi-tenant operations platform featuring **RBAC**, **immutable audit logging**, **fleet-wide diagnostics**, **process and container lifecycle management**, **AI-assisted log/telemetry analysis**, and containerized deployment via **Docker**, **Cloudflare Tunnel**, **Nginx**, and **Vercel**.

RescueMind | *FastAPI, Go, gRPC, RAG, Redis, WebSockets* — IET NITK Project Expo 2026
https://github.com/Jishnu-Prasad888/RescueMind

- Built an AI-powered disaster intelligence platform for real-time satellite and aerial imagery analysis using CNN-generated grid heatmaps across 5 disaster categories.
- Engineered a distributed backend using FastAPI, Go, gRPC, Redis, and WebSockets to process live RSS feeds and AIS ship-tracking streams with end-to-end response latencies of ~**50–200 ms**.
- Designed low-latency event-driven system architecture supporting concurrent multimodal data ingestion, inference, and real-time dashboard updates.
- Built an autonomous rover prototype using YOLO-based vision and sensor fusion with a hardware safety override.

PacketLens | *C++, Qt6, Python, Linux Networking, MCP (Model Context Protocol)*
https://github.com/Jishnu-Prasad888/PacketLens

- Built a **real-time network monitoring tool** with packet capture, flow aggregation, and process-level attribution on Linux.
- Designed a dual-interface system with flow tables and interactive graphs for network visualization.
- Highlighted high-bandwidth processes using node scaling and protocol-based color coding.
- Mapped live network connections to originating processes for system-level visibility into traffic behavior.
- Exposed a lightweight HTTP API and MCP integration for natural-language querying of network activity.

Browser-RAG — Local Browser History Retrieval-Augmented Generation | *Python, FastAPI, React, TypeScript, ChromaDB, Ollama, LLaMA 3.2*
https://github.com/Jishnu-Prasad888/Browser-RAG

- Built a local-first RAG system that extracts and processes browser history from Chrome, Firefox, and Edge across Windows and Linux.
- Designed a content pipeline to download, chunk, and embed visited web pages into ChromaDB for efficient semantic retrieval.
- Implemented a FastAPI backend with Ollama-powered LLaMA 3.2 for context-aware question answering over personal browsing data.
- Developed a React + TypeScript chat interface with conversation history, markdown rendering, and responsive UI.
- Integrated smart filtering, deduplication, and domain blocking to optimize storage and retrieval quality.

Technical Skills

Languages: Python, C++, Go, JavaScript/TypeScript *Familiar:* Java, Bash, GraphQL, MATLAB, R
Frameworks: FastAPI, Django, React, Next.js
ML & CV: Langchain, MCP(Model Context Protocol), ADK (Agent Developemnt Kit), PyTorch, TensorFlow, OpenCV
Tools: Linux, Docker, Kubernetes, PostgreSQL, Redis, Git

Leadership & Activities

Top 10 — Caterpillar Autonomy Challenge IIT Madras
Team Member, NITK CATERBOTS

- Designed and built an autonomous Mars-terrain rover capable of crater detection, sand berm construction, and NLOS navigation out of 1,922 competing teams.

IET NITK Student Chapter NITK Surathkal
Project Lead — RescueMind & Web Team member

- Led a team building RescueMind, a disaster response system with routing, CNN-based classification, and RAG-based advisories.

Amateur Astronomy Club (AAC) & SMILE Club NITK Surathkal
Chief Technical Coordinator in AAC and Events & Publicity Coordinator in SMILE

- Co-led Arduino-based astronomy projects including stellar mass estimation and ML-based multi-camera video stitching.
- Organized technical events, secured sponsorships, and managed sub-teams for structured event delivery.